LECTURE-READY • ANIMATED

VGG16

How 3×3 filters build a deep visual hierarchy



Oxford Visual Geometry Group • original VGG family (2014)

The entire idea in three operations

Follow a single image from pixels to a class prediction.

FEATURE EXTRACTION

Convolution

Learn local patterns using small 3×3 filters. Multiple filters create multiple output channels.

DOWNSAMPLING

Max pooling

Keep important activations while reducing the spatial resolution by 2×.

DECISION

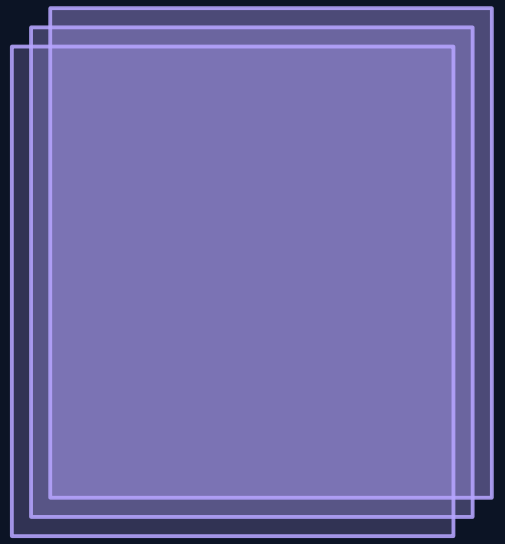
Classification

Combine high-level features and predict a class using learned dense layers.

Repeated blocks: CONV → CONV → POOL → ... → CLASSIFIER

Start with a 224 × 224 RGB image

Why does VGG16 receive three channels?



224 × 224 × 3

The three channels are R, G, B

RED channel

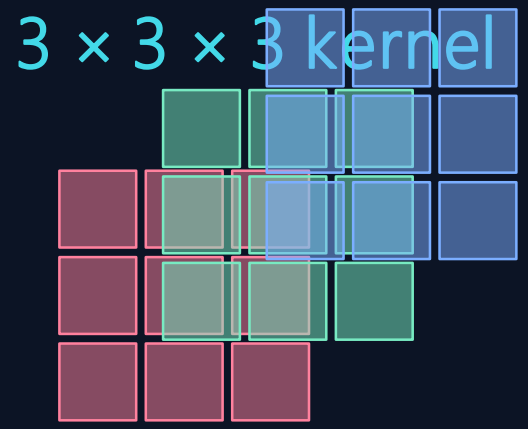
GREEN channel

BLUE channel

224 × 224 = pixel positions • **3 = channels per position**

A 3×3 filter is NOT just a flat 3×3 grid

The kernel extends through all input channels.







one map

224 × 224 × 1

One filter

→

one output feature map

VGG applies 64 DIFFERENT learned filters to produce 64 output maps.

Animation: watch the 3×3 filter slide

The animation loops automatically in PowerPoint Slide Show.

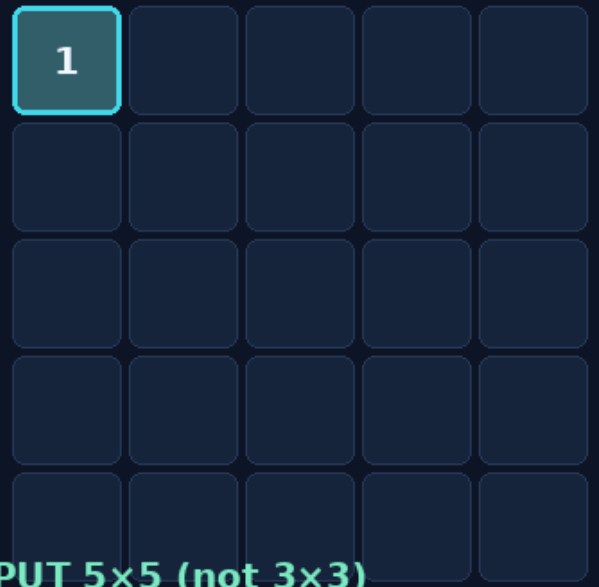
3×3 FILTER SCANS THE INPUT

Example: padded 5×5 image with a learnable 3×3 filter (mean weights shown here).

INPUT + 1-PIXEL PADDING

0	0	0	0	0	0	0
0	1	1	2	1	0	0
0	1	5	7	2	0	0
0	0	6	9	7	2	0
0	1	3	8	6	1	0
0	0	1	4	3	0	0
0	0	0	0	0	0	0

OUTPUT FEATURE MAP



AT THIS POSITION

8 ÷ 9 ≈ 1

Row 1, Col 1

Step 1 / 25

ZERO PADDING KEEPS THE SPATIAL OUTPUT 5×5 (not 3×3)

Why does the depth become 64?

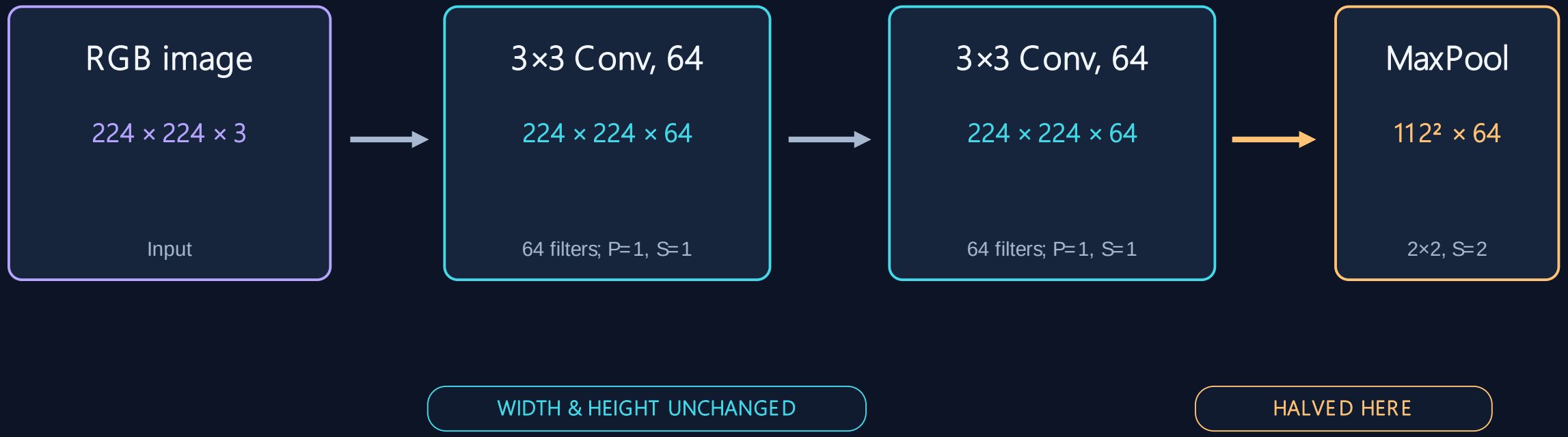
Output depth = number of convolution filters, not the RGB input depth.



64 different filters → 64 learned feature maps at every pixel location

In VGG16, two convolutions come first

Both use stride 1 and 1-pixel padding, so image size is preserved.



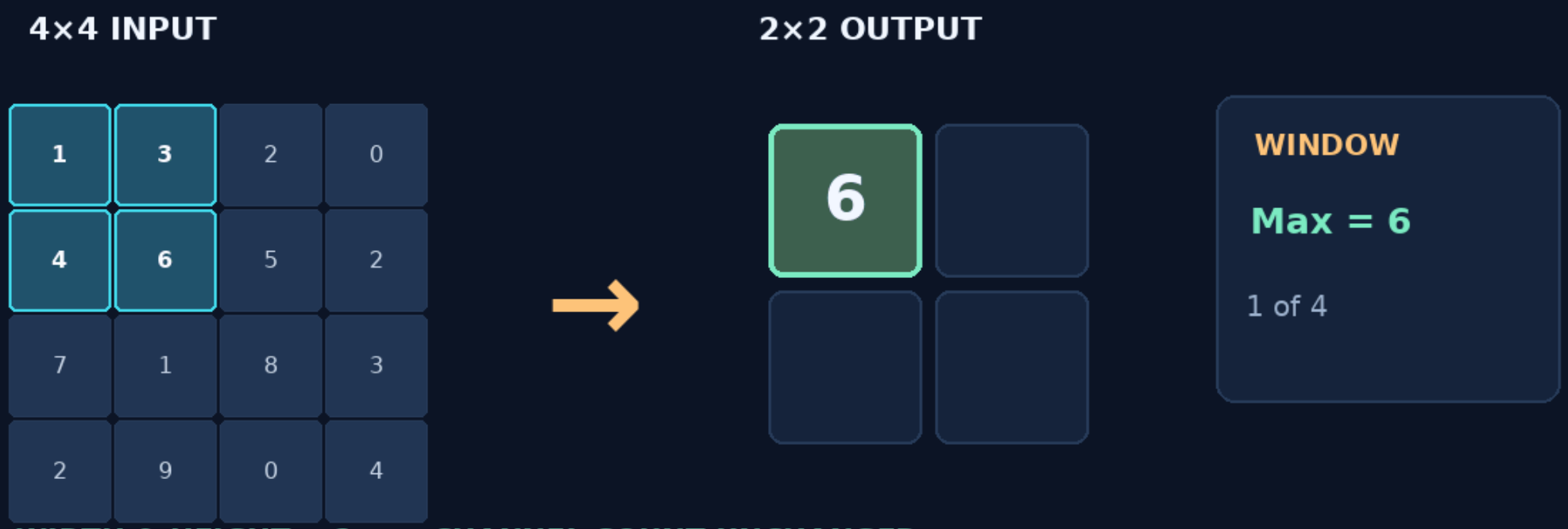
The first Conv has 3 input channels; the second Conv has 64 input channels.

Animation: max pooling picks the maximum

A 2×2 window with stride 2 halves width and height.

2×2 MAX POOLING • STRIDE 2

Keep the largest activation in each non-overlapping 2×2 window.



One formula explains both operations

Spatial dimension is controlled by kernel, padding, and stride.

$$\text{Output size} = \lfloor (\text{Input} + 2\times\text{Padding} - \text{Kernel}) / \text{Stride} \rfloor + 1$$

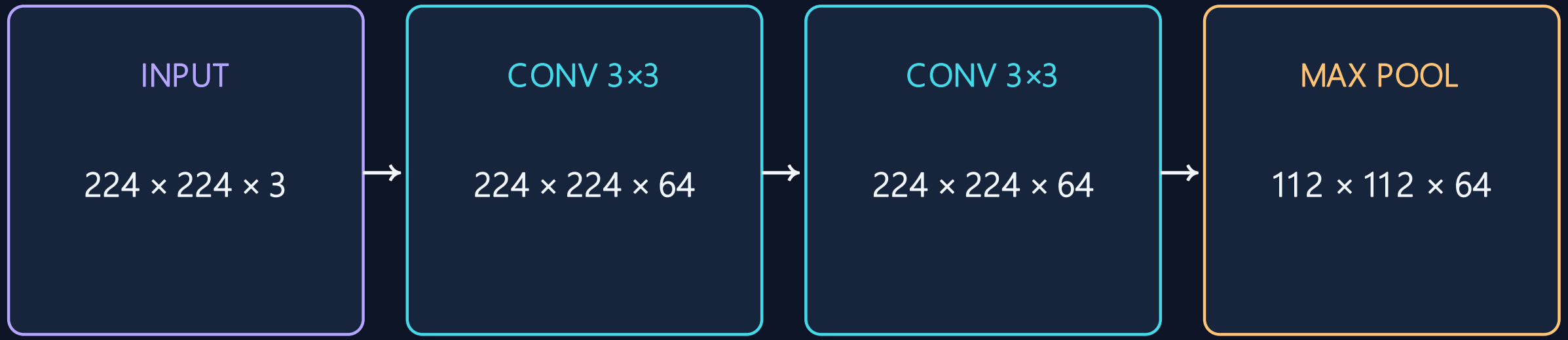
Conv: $(224 + 2 - 3) / 1 + 1 \rightarrow 224$

Pool: $(224 - 2) / 2 + 1 \rightarrow 112$

The depth changes ONLY when the number of convolution filters changes.

Checkpoint: complete VGG16 Block 1

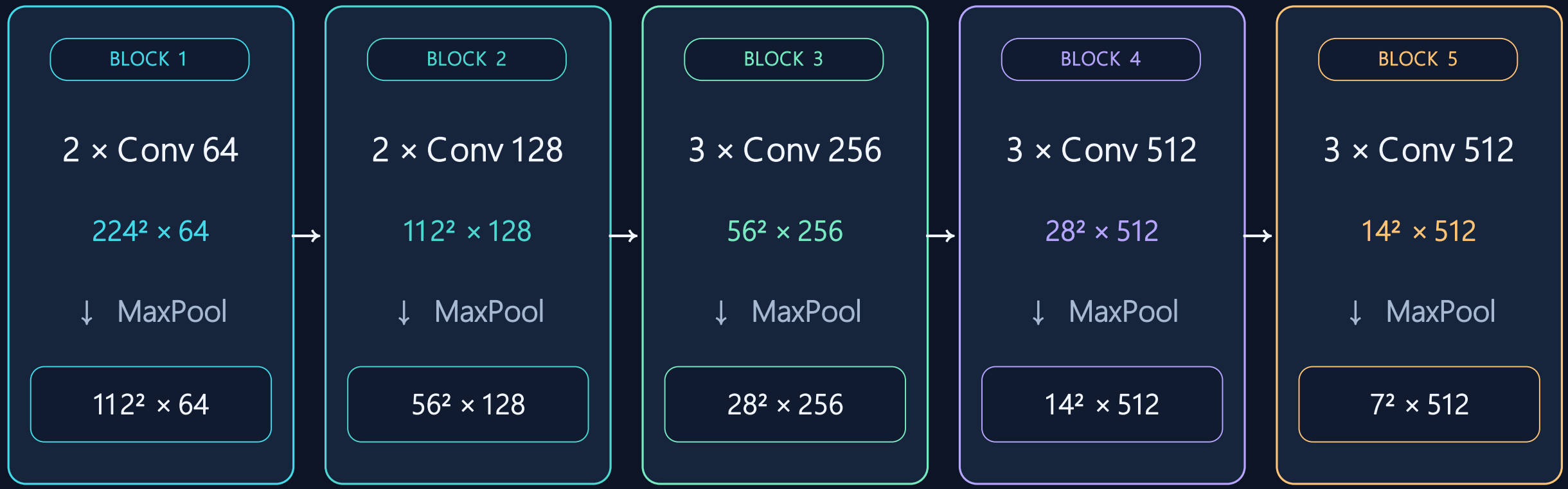
Follow the output tensor shape after every operation.



The first two steps add / learn features; pooling reduces spatial resolution.

VGG16: all five convolution blocks

As width and height decrease, the feature depth generally increases.



Then: 7 × 7 × 512 → Flatten → FC 4096 → FC 4096 → FC 1000

VGG16 layer-by-layer shape guide

All 3×3 convolutions use padding 1 and stride 1.

STAGE	CONVS	FILTERS	SIZE AFTER CONV	SIZE AFTER POOL
Block 1	2	64	224 × 224	112 × 112
Block 2	2	128	112 × 112	56 × 56
Block 3	3	256	56 × 56	28 × 28
Block 4	3	512	28 × 28	14 × 14
Block 5	3	512	14 × 14	7 × 7

Tensor depth after each stage is the filter count shown in column 3.

Why increase the number of channels?

Deeper layers can represent more varied visual patterns.

64 FILTERS

Early layers

High-resolution grids capture local contrast, edges, and color transitions.

128–256 FILTERS

Middle layers

Lower-resolution maps combine local features into textures, corners, and parts.

512 FILTERS

Deep layers

More channels can respond to larger, more complex visual configurations.

Smaller spatial grids ≠ fewer features. Depth and spatial size are different axes.

Why VGG prefers many small 3×3 filters

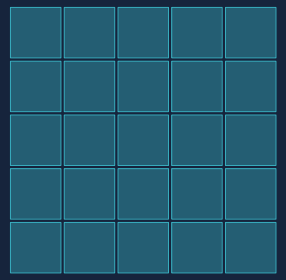
Stacked 3×3 layers expand the effective receptive field.

ONE 3×3



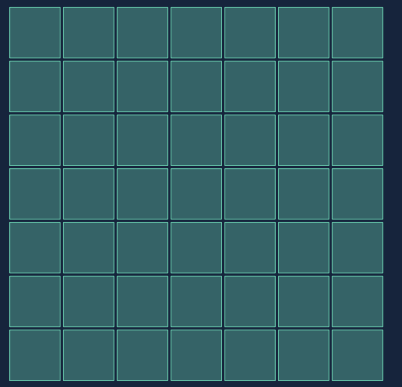
3 × 3 receptive field

TWO 3×3



5 × 5 receptive field

THREE 3×3



7 × 7 receptive field

For equal channel widths: 3 stacked 3×3 layers use $27C^2$ weights vs $49C^2$ for one 7×7.

What does “16” in VGG16 mean?

It counts the layers with learnable weights.

13

convolutional layers

+

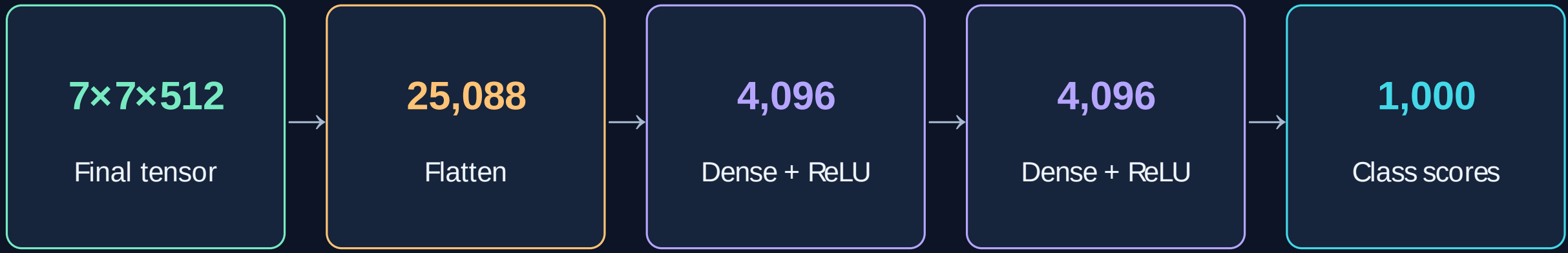
3

fully connected layers

5 max-pooling layers are present, but are NOT counted in the “16”.

From convolutional features to classification

The original VGG16 ImageNet head uses three dense layers.



The 1,000 classes belong to the original ImageNet setup; replace the head for a new task.

Does VGG16 use backpropagation? Yes.

Convolution filters are learned, not hand-designed.

1. Forward pass

Image → filters → prediction

2. Compute loss

Compare prediction with label

3. Backpropagation

Find gradients for learnable weights

4. Optimizer update

Adjust convolution and dense weights

Backprop changes learned filter weights; max pooling itself has no trainable parameters.

Three misconceptions to eliminate

Ask students whether each statement is true or false.

"3×3 convolution makes 224 become 222."

FALSE • With padding 1, stride 1, 224 remains 224.

"64 filters produce 64 feature maps."

TRUE • Each output channel comes from one filter.

"Max pooling changes 64 channels into 32."

FALSE • It halves width and height, not channels.

Student challenge: predict the tensor shapes

Give students 45 seconds before showing the next slide.

Input 112 × 112 × 64 → Conv3×3, 128 filters, P=1, S=1

What is the output tensor shape?

A 110 × 110 × 64

B 112 × 112 × 128

C 56 × 56 × 128

D 112 × 112 × 64

Answer: 112 × 112 × 128

The core rules you should remember about VGG16.

B • 112 × 112 × 128

3×3 convolution, P=1, S=1	Preserves width and height
Number of filters	Sets the new channel count
2×2 max pool, S=2	Halves width and height, keeps channels
VGG16	13 convolutional + 3 fully connected layers